HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained embedded C/C++ software for resource-constrained microcontrollers, focusing on bare-metal and RTOS environments to support physiological monitoring solutions.
- Designed and implemented drivers for embedded peripherals including **UART**, **SPI**, and **I²C**, ensuring reliable data acquisition for sensor inputs.
- Debugged complex embedded software issues using hardware debuggers and logic analyzers, identifying root causes and implementing robust fixes within tight development cycles.
- Contributed to the design and development of modular software components, participating actively in code reviews and collaborative technical discussions to ensure code quality and maintainability.
- Investigated and resolved software defects reported from testing phases and early-stage hardware prototypes, implementing corrective actions and updating design documentation.
- Created and maintained comprehensive software documentation, including requirements specifications, design documents, and test records, adhering to strict development process standards.
- Participated in software risk analysis activities, identifying potential hazards and implementing mitigation strategies to ensure the safety and reliability of medical device software.

SKILLS

Embedded Systems – Embedded C, Embedded C++, Microcontrollers, RTOS, Bare-metal, UART, SPI, I²C, ADC, PWM, STM32

Development Practices – Modular Design, Source Control, Code Reviews, Unit Testing, Integration Testing, Software Requirements, Software Design, Risk Analysis, IEC 62304

Tools & Technologies – Python, C, C++, Hardware Debugging Tools, Software Debugging Tools, Git, Jenkins

Domain Knowledge – Physiological Monitoring, Medical Devices, Healthcare Technology, Clinical Decision Support

Programming Languages – C, C++, Python

PROJECTS

Real-Time Physiological Data Acquisition Module

- Developed a C-based module for real-time acquisition and initial processing of physiological data streams from multiple embedded sensors.
- Implemented efficient data buffering and error handling mechanisms for sensor inputs, managing data integrity under varying signal conditions.
- Integrated the module with a simulated RTOS environment to test performance and resource utilization, ensuring adherence to strict latency requirements.

Embedded Software Debugging & Automation Tools

- Created Python scripts to automate the generation of test vectors and to parse raw debug logs from embedded hardware, reducing manual analysis time by 40%.
- Designed and implemented a basic framework for unit testing embedded C functions, including mock objects for peripheral interactions.
- Developed diagnostic tools to aid in the identification of common hardware interface issues (SPI, I²C) on prototype boards.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)